

Report on Resolving libgamemodeauto.so.0 Preloading Errors in Lutris with UE4 Games

The error message "ERROR: ld.so: object 'libgamemodeauto.so.0' from LD_PRELOAD cannot be preloaded (wrong ELF class: ELFCLASS64): ignored." commonly arises in the Lutris gaming platform, particularly when running games via Wine, including those built with Unreal Engine 4 (UE4). This report provides a detailed analysis of the error, its causes, and potential solutions, drawing from forum discussions, GitHub issues, and related documentation. The information synthesized aims to allow readers to both understand the problem and correctly implement a fix where others have failed. The report will focus on the most relevant and reliable information to address the error effectively.

Understanding the Error

The error message indicates a failure to preload the libgamemodeauto.so.0 library. This library is part of Feral Interactive's Gamemode, a system-level utility designed to optimize game performance by temporarily applying performance-enhancing tweaks to the operating system ([Feral Interactive's Gamemode](#)). The LD_PRELOAD environment variable is used to specify shared libraries that should be loaded before others when a program is run ([LD_PRELOAD Documentation](#)).

The core of the issue lies in the "wrong ELF class" component of the error. ELF (Executable and Linkable Format) is a standard file format for executables, object code, shared libraries, and core dumps in Unix-like systems. The "ELFCLASS64" designation implies a 64-bit library, and the error arises when there is a mismatch between the architecture of the library and the process attempting to load it. Common causes include:

- 1. Architecture Mismatch:** A 32-bit process attempts to load a 64-bit library, or vice versa. This is a frequent issue in environments like Wine, where 32-bit Windows games are run on 64-bit Linux systems.
- 2. Incorrect Library Path:** The LD_PRELOAD variable points to a non-existent or incorrect path for the library, or does not exist.

Diagnosing the Root Cause

To effectively address the error, it's essential to accurately diagnose the root cause. Several steps can be taken:

1. **Verify Gamemode Installation:** Ensure that Gamemode is correctly installed on the system. This can typically be done using the system's package manager ([Forum Discussion](#)).

```
```bash
```

## Example for Debian/Ubuntu

```
sudo apt update sudo apt install gamemode ```
```

2. **Check Library Paths:** Confirm the correct paths for `libgamemodeauto.so.0` on your system. Use the `ls` command combined with `find` or `locate` to identify the library's location. Ensure both 32-bit and 64-bit versions are present if running 32 bit Windows games on a 64 bit system.

```
bash sudo updatedb #Update database so the locate command works accurately locate libgamemodeauto.so.0
```

3. **Inspect Lutris Configuration:** Check the Lutris game configuration for any custom `LD_PRELOAD` settings. These settings may be overriding the system defaults and causing the error. Modify or remove incorrect entries in the Lutris game configuration ([Lutris Configuration Guide](#)).
4. **Wine Architecture:** Determine if the game is running in a 32-bit or 64-bit Wine environment. This will dictate whether the 32-bit or 64-bit version of `libgamemodeauto.so.0` needs to be preloaded. This can be checked within the Lutris configuration for the specific game.

## Potential Solutions

Once the cause is identified, the following solutions can be implemented:

1. **Correct LD\_PRELOAD Paths:** Modify the `LD_PRELOAD` variable to point to the correct library path, considering the architecture of the Wine

environment. If using Lutris, this can be set within the game's configuration. It may be necessary to specify different paths for 32-bit and 64-bit libraries. An example is:

```
bash LD_PRELOAD="/usr/lib/libgamemodeauto.so.0:/usr/lib32/libgamemodeauto.so.0"
```

2. **Remove Conflicting Entries:** If the error arises from conflicting or incorrect LD\_PRELOAD entries, remove them to allow the system to use the default Gamemode configuration.
3. **Install Missing Libraries:** If the required 32-bit or 64-bit versions of libgamemodeauto.so.0 are missing, install them using the system's package manager. This may involve enabling multi-arch support on certain distributions.

```
```bash
```

Example for Debian/Ubuntu to enable multi-arch and install 32-bit libraries

```
sudo dpkg --add-architecture i386 sudo apt update sudo apt install libgamemodeauto0:i386 ```
```

4. **Address NV-GLX Error:** Simultaneously, users reported "Xlib: extension "NV-GLX" missing on display" ([Forum Discussion](#)). This issue can occur when the correct NVIDIA drivers are not properly installed or configured. Make sure the correct NVIDIA drivers are installed for your system.

```
```bash
```

## Example for Debian/Ubuntu

```
sudo apt install nvidia-driver- ```
```

Restart your system after installing the drivers. After the reinstall, check your `Xorg.conf` file located at `/etc/X11/xorg.conf` and make sure the NV-GLX extension is properly enabled within the appropriate Module Section.

5. **Flatpak Considerations:** If Lutris is installed via Flatpak, the standard library paths may not be directly accessible. In such cases, it may be necessary to ensure that Gamemode is correctly integrated into the Flatpak environment or to use Flatpak overrides to provide the correct library paths ([GitHub Issue](#)).

```
```bash
```

Example for Flatpak override

```
flatpak override --env=LD_PRELOAD="/app/extra/lib/libgamemodeauto.so.0" net.lutris.lutris ```
```

6. **Runner Compatibility:** Ensure that the Wine runner used by Lutris is compatible with Gamemode. Try different Wine runners (e.g., Lutris-Wine, Wine-GE) to see if the issue is specific to a particular runner version. Runners can be managed and updated via the Lutris interface.

Gamemode Not Starting Issue

Another reported problem involves Gamemode not starting at game launch ([GitHub Issue](#)). Even when Gamemode seems to be running correctly outside of Lutris, the Lutris logs still display errors concerning `libgamemodeauto.so.0`. In this scenario, the following steps can be taken:

1. **Verify Gamemode Status:** Check the status of Gamemode using `gamemoded -s` and `systemctl --user status gamemoded` to confirm it is active.
2. **LD_PRELOAD Workaround:** As a workaround, manually set `LD_PRELOAD` to `/usr/lib/libgamemode.so.0` (or the equivalent path on your system) within the Lutris game configuration. Although the logs may still show errors, this has been reported to enable Gamemode effectively.

3. **Systemd User Instance:** Ensure that the Gamemode daemon is running as a user instance. Enable and start it using:

```
bash systemctl --user enable gamemoded systemctl --user start gamemoded
```

4. **Reinstall Gamemode:** Completely remove and reinstall Gamemode to ensure all components are correctly installed and configured.

Gstreamer Errors

In some cases, the `libgamemodeauto.so.0` error may appear alongside GStreamer-related errors, particularly those indicating "wrong ELF class". These GStreamer errors often stem from architecture mismatches similar to the Gamemode issue. Ensure that the correct 32-bit GStreamer libraries are installed alongside the 64-bit versions if running 32-bit applications with Wine ([Forum Discussion](#)). Typical missing libraries include: `libgstogg.so`, `libgstflxdec.so`, `libgstautoconvert.so`.

```
```bash
Example for Debian/Ubuntu to install 32-bit GStreamer libraries
sudo apt update
sudo apt install gstreamer1.0-plugins-good:i386 gstreamer1.0-plugins-bad:i386
```
```

Summary

The "ERROR: ld.so: object 'libgamemodeauto.so.0' from LD_PRELOAD cannot be preloaded (wrong ELF class: ELFCLASS64): ignored." error in Lutris, especially with UE4 games, typically results from architecture mismatches, incorrect library paths, or misconfigured environments like Flatpak. By methodically diagnosing the root cause and applying the appropriate solutions—such as correcting LD_PRELOAD paths, installing missing libraries, addressing Flatpak-specific issues, and ensuring Wine runner compatibility—users can effectively resolve the error and optimize their gaming experience. The additional step of ensuring correct NVIDIA

driver installation and configurations will help resolve graphic extensions issues like NV-GLX, where present.

- Verifying Gamemode status and using the manual LD_PRELOAD workaround can sometimes bypass persistent logging errors and enable Gamemode functionality.
- Ensure that if the reported errors appear alongside GStreamer errors, GStreamer libraries are installed and configured correctly as well.

References

Feral Interactive's Gamemode. <https://github.com/FeralInteractive/gamemode>

LD_PRELOAD Documentation. <https://man7.org/linux/man-pages/man8/ld.so.8.html>

Forum Discussion. (2020, November 26). League Of Legends Gamemode ERROR - Support - Lutris Forums. <https://forums.lutris.net/t/league-of-legends-gamemode-error/11071>

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GitHub Issue. (2019, July 28). [Flatpak] 'libgamemodeauto.so' from LD_PRELOAD cannot be preloaded · Issue #2248 · lutris/lutris · GitHub. <https://github.com/lutris/lutris/issues/2248>

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